

Institute of Actuaries of India

Subject SA2 – Life Insurance

March 2021 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable

Solution 1:

i) Before accepting the offer, I would consider the following factors:

- I am eligible to be an Appointed Actuary as per Reg 3B of IRDAI(AA) Regs, 2017 which suggests that –
 - I am Fellow of Institute of Actuaries with the specialization in the subject
 - Total experience of 10 years in the life insurance industry of which 5 years must be post the qualification
 - There must be 3 years spent in the annual statutory valuation of a life insurer post the qualification [1.5]
- I must have a certificate of practice (COP) issued by IAI [0.5]
- I have appropriate knowledge and practical experience relevant to the upcoming role [0.5]
- I possess and demonstrate proper professional behavior and standards [0.5]
- In line with the previous experience, whether I will be able to manage the responsibilities on my own or do I need to have a recourse to an actuary who possess such knowledge and relevant experience [0.5]
- I have confidence in the way the Company under consideration is conducting its business and its practices [0.5]
- There shall be a contractual relation between the Company and me as employer employee and the same should be enforced by the appointment letter [0.5]
- The Company shall provide access to senior management, required data, access to any information likely to affect the financial condition of the Company [0.5]
- Also the reason why the previous actuary left might be of relevance as well? It would help to understand if there were any professional issues I should be aware of before taking up the assignment? [0.5]
- What lines of business does this company sell (predominantly UL/par; Group/ Individual) and do I have enough experience in that kind of business [0.5]

[Max 5]

ii) In case I have worked with the Company before, then in addition to the above-mentioned points, the following shall also be important to consider:

- Could there be any **professional conflict** between the Appointed Actuary role and the management of the Company
- Could there be any **undue influence** of any of the members of the Company which could possibly impact the way Appointed Actuary is supposed to discharge its liabilities
- Are there any known **areas of disconnect** in the way the Company is run or managed and can it be ensured that none of the past events impact the functioning of AA way forward
- Are the current practices of the Company acceptable to what you consider as professional standards (say reserving method/ assumption setting, solvency evaluation, ALM, product pricing). Professionally if you are not aligned, it might be better to express that to the management before accepting the role.
- Are there any disagreements or discord with the regulators on certain business practices of the Company? Any pending notices pertaining to previous years, etc.

[1 marks each]

[Max 3]

- iii) The reserving and solvency of a life insurance company is governed by various regulations, guidelines or circulars issued by IRDAI and IAI, primarily IRDAI(ALSM) Regulations, 2016, Section 64V & VA of the Insurance Act, 1938, APS-2 and APS-7 as issued by IAI time to time.

Reserving:

Methodology:

- For the long term business under Individual and Group platform, reserves are to be calculated using Gross Premium Valuation (GPV) methodology for every policy on the books
- For the one-year renewable group business, the reserves shall allow for Unearned premium (UPR), Premium deficiency and Incurred but not reported claims.
- For the riders attached to one-year renewable group term business can also follow the same methodology as above
- For the other riders, the reserves shall be higher of GPV or UPR
- The valuation methodology shall take into account all possible contingencies under which premiums or benefits may be payable over the policy tenure
- The GPV methodology shall allow for:
 - Premiums and benefits payable under the contract including basic as well as rider benefits, vested bonuses, future bonuses
 - Commission payable and expenses to be incurred under the contract
 - Allocation of profit to shareholders
 - Any additional cost of guarantees and options
 - Reinsurance cashflows, if applicable
- The reserves at all times, should be held such that they are as high as Guaranteed Surrender Value and not be less than 0
- Since the Company writes Unit linked business as well, for this business, unit and non-unit reserves shall be held separately. Neither the unit nor the non-unit reserve in respect of a policy shall be negative
- For the participating business, it is important to consider past practice, policyholder's reasonable expectations, the extent by which liability can be reduced by decreasing the future bonus rates and the principles suggested by GN22
- In addition to the base reserves calculated as per the methodology above, there would be additional reserves held for:
 - Incurred but not reported claims
 - Revival reserve in respect of the policies in the lapse status
 - Closure to New business reserves in accordance with the provisions of APS-2
 - Options and Guarantees
 - Policies in respect of which extra premium has been charged
 - Mismatch reserve for Unit linked business should unit liabilities are not matched with corresponding assets

[Max 7]

Assumptions:

- The valuation parameters such as mortality/morbidity, expenses, persistency, valuation rate of interest shall be determined based on the actual underlying experience of the insurer, wherever available and adjusted for the suitable MAD in accordance with the provisions of APS-7
- The mortality/morbidity assumptions shall refer to a standard table and the assumption rate should be determined such that it reflects the actual experience, if available
- The expense assumption should be based on actual experience and the inflation rate to be consistent with this expense assumption
- The valuation interest rate should be determined on prudent assumptions of the underlying yields, with an allowance for risk and expenses. For the non-participating business, the risk of a decline in the interest rate should be duly recognized and a suitable adjustment shall be made to the reserves
- The persistency/lapse assumptions should be based on the actual experience and consider products, target market, distribution channels etc.

[Max 3]

Solvency:

- Every insurer at all times is required to maintain its Available Solvency Margin (ASM) at a level not lower than one hundred and fifty percent of the Required Solvency Margin (RSM)
- While calculating ASM, the assets are to be valued in accordance with the Preparation of Financial Statements and Auditor's Report of Insurance companies, Other Forms of Capital and investments, as applicable from time to time
- There are certain assets to be placed zero value while calculating ASM, which are:
 - Agents and Intermediaries balances and outstanding premium in and/or outside India
 - Sundry debts, advances, receivable not expected to be realized
 - Furniture fixture, dead stock, stationery
 - Deferred expenses
 - Debit balance of P&L and any other fictitious assets
 - Reinsurers balance outstanding for more than 90 days
 - Leasehold improvements
 - Service Tax unutilized credit outstanding or more than 90 days
 - Any other asset admissible as per Section 64V of the Insurance Act, 1938
- For the calculation of RSM, the factors as per IRDAI (Actuarial Report & Abstract) 2016 shall be followed.
- There are two factors – one to be applied to the reserves and the other to be applied to the sum at risk. The factors vary by the individual/group, life/pension/annuity/health, linked/non-linked, participating/non-participating, with/without guarantees.
- There are separate factors applicable for health and other than health riders.

[Max 6]

iv) Covid-19 has impacted different countries differently and its full impact on any country or region is still evolving. The same is true for India where significant different impacts have been experienced across various regions. In this scenario, it might be really difficult to be able to come up with close enough approximation for the likely impact of the pandemic, however, the following considerations might be made:

- **Likely impacted areas:** As a life insurance company, most of the liabilities shall be triggered by the likely increase in mortality rate primarily. But since the company also writes health insurance business, it might have an impact due to the increased incidences of hospitalization or subsequent higher incidences of health complications. So it is important to understand where all it might impact.
- **National statistics:** It is important to keep a tab on the national data suggesting the infection rate as well as the fatality rate due to the pandemic. It might not be directly relatable to the Company's portfolio but might be a useful indicator, given the lack of sufficient data on this pandemic yet
- **Reinsurers view:** The view of Reinsurers may be considered as they deal with a larger data set compared to one Company and may be able to estimate numbers in a more effective manner.
- **Actual claim analysis:** Since it's been almost a year since the breakdown of the virus in the country, it might be useful to draw some insights based on the actual claims experience so far. The trend analysis might be useful to carve out any abnormalities seen in any of the portfolios this year. The data can be split by age/co-morbidities/type of product/premium size etc. to understand the underlying. To be able to do these analysis, it is important to capture all this information at underwriting and/ or claims stage. Ensure the IT system has the capacity to store this info and the front end staff is sensitized about its importance.
- In addition, there might be larger delay in intimating claims due to lockdown or restricted movements, so a **higher IBNR** than usual might be allowed for while analyzing claims
- Due to the co-morbidity issue, the cause of death may also be debatable. Has the patient died of lung / liver infection (which might have happened due to Covid-19 complications)? Quality of the data is critical as well
- There could also be higher claims due to sustained lock down, people losing jobs, not interacting with friends/ family, prone to depression and other mental health conditions.
- The disease has impacted various geographic regions differently. It is important to correlate this regional experience with the business volume written in these areas.
- There may be a positive impact of the pandemic as well, where in people are now taking good care of their health and making lifestyle changes to lead a healthier life.
- It may be a good idea for the company to provide some app to all its policyholders to monitor and engage with him to prevent claims. There will be some cost involved in such initiatives though. It will help monitor how your policyholders in vulnerable ages or with comorbid conditions can be managed

[½ marks for each point, Max 5]

Should there be any additional reserves required might depend on:

- The insights based on the above analysis – whether it suggests adverse impact or not
- It is also important to understand that this is a one-time pandemic which will go away after the vaccines are made available; which will not have any major long term effect on the mortality, except for a spike for 2-3 years. However, if it is meant to cause a long term effect, that needs to be taken care of.
- The existing level of prudence in the mortality assumption also plays a role in deciding whether an additional reserve would be required. Reinsurers interpretation of this can be crucial, as they will have a wider experience.
- If found appropriate, the Appointed Actuary should hold reserves separately for Covid-19 claims
- Even if the existing reserve seems prudent enough, AA may choose to set aside some extra reserves for any adversity which might arise in the near future
- The reserve could be held as a certain % of claims expected over and above the normal levels or a higher MAD might be allowed for and/or higher IBNR might be incorporated in the reserve calculations

[1/2 marks for each point, Max 3]

[8]

v) The bonus declaration shall take into account the following:

- **actual returns on the underlying assets:** it is important to understand the impact of falling interest rates on the returns of the underlying fund. The ongoing interest rates shall impact different cohorts of policies differently, such as the policies written recently where most of the premiums are yet to be invested might be impacted the most and the policies closer to maturity where most of the premiums are already invested might be impacted the least. So as a first step, it shall be very important to split the policies into the reasonable cohorts and look at the impacts separately.
- **bonus supportability of the different cohorts:** next, it will be important to understand the interest rate outlook if the ongoing fall in interest rates is temporary or not and whether the future outlook is optimistic or pessimistic. Some discussions and insights from the Investment team shall be helpful. Based on the identified cohorts and the actual returns so far, a bonus supportability study should be carried out under various future interest rate scenarios.
- **past practice adopted by the Company while declaring bonus rates:** the past practice followed in such scenarios shall also be considered in line with the 'Principles and Practice of Financial Management' (PPFM) of the Company.

- **policyholder's reasonable expectations (PRE):** PRE might be based on the how the benefit illustrations are issued, what does bonus letter or any other communication with the policyholder seem to suggest, what is the practice adopted by the competitors, recent year's bonus rates, PPFM etc. A reasonable policyholder shall expect a return in line with the market scenario and hence, a reduction if required based on the current scenario might be considered appropriate.

In addition, the Company may also consider if any revision is required in the current bonus rates being shown in the Benefit Illustrations, so that the PRE is set in an appropriate manner.

Level of 'Funds for Future Appropriation' (FFA): the current level of FFA, especially contributed by the cohorts where bonus rates are to be declared shall also be an important factor while deciding on this year's bonus rates. Some part of FFA might be used to assist the deficiency in the bonus rates, however, the same shall be dealt with caution as FFA is usually accumulated over long time and hence, it should not be utilized for a limited set of policies. In addition, if the Company cannot infuse capital under the participating fund (assuming the Company is in existence for more than 12 years), the role of FFA shall be critical to be able to write future new business by funding new business strain.

[Max 10]

vi) As per the IRDAI (Non Linked Insurance Products) Regulations, 2019, the With Profit Committee Report is expected to cover the following with respect to the bonus declaration, along with the other particulars:

- Bonus Earning capacity including its calculations
- Sensitivity analysis of bonus rates and basis
- How PRE is met
- Treatment of FFA

[2]

Hence, in accordance with the above, the WPC Committee would be expected to be involved significantly in the entire process of the bonus declaration starting from the calculations of the asset share and the bonus earning capacity. The Committee would be expected to understand the approach, methodology and be satisfied with the outcome and the recommendation. It would be expected to take a holistic view while reviewing the with-profit business. In addition, the Committee should also evaluate how any decision would impact not only existing policyholders but also the future policyholders and whether they are being treated fair and equitable.

[3]

The WPC Committee should put forward its view points and share its agreement/dis-agreement with the Company. It may also suggest an alternative approach, if any to the overall process. It is expected that the differences if any between the Committee and the Appointed Actuary are resolved before the final recommendation is made on with profit business. However, if any differences remain unresolved, the same should be included in the final Committee report. The report needs to be placed with the Board as well as to be sent to IRDAI.

[2]

Since the AA retains entire responsibility of his or her work, he/she retains the final say on whether or not any element of his/her work needs to be changed as a result of the Committee report.

[1]

[8]

Solution 2:

i) The ALM policy approved by the board as per the ALM circular issued by the IRDAI should highlight

- the ALM framework of the company considering the asset-liability relationships,
- the insurer's overall risk tolerance,
- risk and return requirements,
- solvency position and
- liquidity requirements.

[Max 2]

The key components of ALM framework are as follows.

1. List of the ALM risks that the company is exposed to. The risks highlighted in ALM circular that a company should consider are
 - a. Market risk (includes equity, interest rate and credit risk),
 - b. liquidity risk and
 - c. underwriting risk. [1]
2. The analysis mechanism of the risks identified to ensure that the impact of ALM risks on key metrics is well understood. [1]
3. Management of ALM risks: The ALM position and the risks should be managed using a range of policies to ensure that the investment activities and asset positions are appropriate to company's liability, risk profiles and solvency positions. [1]

Some of the policies to manage the risk and ALM position are:

- a. Controls in Investment Strategy or Hedging,
- b. Setting risk limits or tolerance limits, [½ + ½]
4. Monitoring and reporting of ALM position on ongoing basis through development of control and reporting procedures. [1]

The ALM policy should also talk about the governance structure of the ALM function. [1] [5]

ii) Following is the list of ALM related risks:

- a. Market Risk is the risk of sudden changes in market values of the equity and other non-fixed income asset classes portfolio thus impacting asset values adversely.
[½ for listing the risk and ½ for explaining]
- b. Interest Rate Risk: is the risk arising from a change in interest rates which may have different impacts on assets and liabilities thus changing the net position in an adverse manner.

[½ for listing the risk and ½ for explaining]

This risk also includes:

- Reinvestment risk: risk that the future proceeds will be reinvested at a lower rate
- Guarantee risk: risk that the minimum guaranteed investment rate will not be met

[1]

- c. Credit Risk: is the risk of economic loss arising from widening of spreads on corporate bonds and counter parties failing to make payments which they are obliged to.

[½ for listing the risk and ½ for explaining]

The risk arises from:

- Creditworthiness of the counter parties
- Concentration risk associated with exposure to a single issuer, promoter group or an industry

[1]

- d. Liquidity Risk: the risk of loss arising from having to sell assets at lower prices (stress prices) to meet surrender obligations in the stress scenario.

[½ for listing the risk and ½ for explaining]

- e. Underwriting Risk is the risk of change in profile of the liabilities due to change in behavior of the policyholders say lapses/surrenders or due to higher mortality experiences which could distort the ALM position of the company.

[½ for listing the risk and ½ for explaining]

[Max 5]

Risk metrics that should be analysed on a regular basis would include, but not limited to, the following:

- a. Solvency ratio
- b. Liquidity ratio (defined as the excess of liquid assets and operating cash flows over customer related cash demands)
- c. Modified/Macaulay duration of assets and liabilities and the duration gap for different lines of business.
- d. Cash flow matching for different lines of business and specific products with inherent guarantees.
- e. Current level of guarantee rates (i.e. the minimum investment return company needs to earn to meet all its guaranteed liabilities) for different products.
- f. Exposure to bonds of different ratings in the fixed income portfolio.
- g. Concentration to a single issuer, promoter group or industry.
- h. Exposure to growth assets in respective funds.
- i. Economic capital requirement for different ALM risks.

[1/2 marks for each point, Max 4]

[9]

iii)

The provisions that are mandated by the regulator to protect the interest of policyholders in Linked Discontinued fund as per the Linked regulations of 2019 are:

- a. Discontinuance fund has to managed as a segregated fund (following all the norms applicable for any segregated fund)
- b. There is a minimum guaranteed return of 4% at the time of exit applicable to the discontinued fund. In case the fund investments don't earn enough; the shareholder has to make good the shortfall.
- c. The excess income earned in the discontinued fund over and above the minimum guaranteed interest rate shall also be apportioned to the discontinued policy fund in arriving at the proceeds of the discontinued policies and shall not be made available to the shareholder
- d. the fund management charge shall not exceed 50 basis points per annum.
- e. Only fund management charges shall be applicable on such funds.
- f. There were prescribed asset classes in which the fund could be invested viz. govt. bonds and money market instruments along with prescribed maximum exposure to each as per the 2013 linked products regulation. However, the same got removed from 2019 linked products regulations.

[1/2 mark for each point]

[Max 2]

iv) Any proposed investment strategy should follow the basic principle of

- To minimize the risk. As the fund is mark-to-market, this could be done by ensuring investments in short duration bonds to reduce volatility of the NAV on interest rate movements in market.
- To maximise the overall return

The proposed investment strategy for discontinuance fund should consider the following:

- a. There is a guarantee of 4% that the investments should meet. In case the guarantee is not met; the shortfall needs to be met by the company.
- b. As per the linked products regulations of 2013, there were investment restrictions in terms of exposure to asset classes (the only assets allowed were govt. bonds and money market instruments) and the maximum limit in each asset class, however, the same got removed from the linked product regulations of 2019. Hence, there is no restriction in terms of asset classes and limit on exposure to particular asset class.
- c. Furthermore, as per the linked products 2019 regulations, there should be only one discontinued fund applicable for all linked life policies which would be having policies sold pre and post 2019 linked products regulations. Hence, the investment restrictions specified in linked products regulations of 2013 will be applicable on the whole linked life fund or not is arguable.
- d. The assets should be matched by nature,
 - because of the hence it might be appropriate to invest in fixed interest instruments, like G- Sec, bonds etc.
 - after meeting the minimum return, they can invest in riskier assets like equities, in order to maximise the overall fund returns
- e. The assets should be matched by term
 - the tenor of assets in discontinued fund should be less than 5 years because of the 5-year lock-in period in discontinuance fund.
 - In case of falling interest rate scenario, short term investments will not be able to support meeting the guarantee.
 - In falling interest rate scenario the reinvestment risk will get exasperated, meeting the 4% even more risky
 - In a rising interest rate scenario the risk of investing long will lead to a ALM mismatch if the interest were to fall. As the longer term assets will fall more than the liabilities
 - While calculating the duration of the liability, it may be important to estimate the expected new money that will come into the fund, as it will have an impact on the outstanding duration.
- f. The minimum investment guaranteed of 4% is applicable only at the time of exit which could be due to death, revival or completion of lock-in period. The minimum guarantee is not applicable to be earned every year. In that case, it is also important to understand the return earned by the policies in the fund over the last few years. And plan the future investment strategy suitably keeping in mind the expected interest rate movement in the balance tenor of the liability.

- g. Since the movement of cashflow can be fluctuating (due to policyholder behaviour) it is important the assets invested are marketable and liquid.
- h. Liquidity may also be managed by future new money coming in the fund due to new discontinuance
- i. The only income to the shareholder in the linked discontinuance fund is through FMC which is limited by the regulator to maximum 0.5% of the fund. So in order to maximise the quantum of return, they have to ensure the fund value can be maximised
- j. Since the only charge allowed is 0.5% of FMC, which includes any cost of investments. So the company has to choose investments where the expense of investment is minimum, like G-Secs, T-Bills.

[1/2 mark for each relevant point]

[Max 6]

v) Below are the implications of the proposal made by the CFO correspond to increase in equity proportion in participating fund:

- a. Basis the investments regulation of 2016 from IRDAI, the life insurance company should invest at least 50% of the life fund (except pension, annuity, group and non-unit reserve of linked business) in the central govt. securities, state govt. securities or other approved securities. Further 15% should be housing and infra bonds. In order to comply with the regulations, the company could have maximum 35% of the fund invested in equity. Further, due to operational ease, companies generally maintain certain amount of fund in cash or cash equivalents as well. By investing 40% in equity asset class, company would be non-compliant to the investments regulation.
- b. Equity investments has the potential of increasing the overall return of the par fund and with that the ability to pay higher bonuses to the policyholder and higher returns to the shareholders as well.
- c. The company has to assess the level of guaranteed benefits that is embedded in their par products, higher the guarantee in the product; the less the company will have investment freedom to invest in risky assets.
- d. With every declaration of bonus, the vested bonus adds to the guarantee in the policy. The company has to assess the minimum return on the fund required to meet these guarantees. To the extent to meet this guarantee, the investments will have to be fixed interest assets (G-Secs and bonds); only any excess can be available for other investments
- e. Investments in equity is considered as risky due to volatility in returns. Coupled with that, the accounting of equity assets happens at market value unlike bonds which are valued at book value. So, any volatility in the equity assets impacts the asset side of the Balance Sheet.
- f. While there is an upside to the equity investments, there is potential downside, this can impact the bonus paying capacity of the company under equity stress scenario.
- g. Also the equity returns could be volatile. But participating policyholders do not like volatility in their bonus rates. They expect smoothing of returns
- h. Increasing the exposure of equity asset class in participating fund would have implications on:
 - Volatility of solvency position of the company

- Assets liability management of the company
- Volatility of shareholder value / embedded value of the company and
- PRE of the policyholder
- i. The volatility in bonus rates could be managed by ensuring stability in the reversionary bonus payouts and keeping the terminal bonus volatile. However, this will have influence of completion and the PRE set by the company through benefits illustration and past experience of the company.
- j. There may also be competitive pressure on how much reversionary bonus can be declared.
- k. Further, it will become increasingly complex to identify which generation of policyholders should get impacted by the equity investment and by how much, hence maintaining equity among policyholders
- l. The implication of increased equity exposure will be on ALM position of the company. There is a need for asset hypothecation to understand which portion of policyholder liability is backed by equity assets and how the cash flow position will change in case of stress scenario.
- m. The volatility to future bonus rates would make the shareholder transfers equally volatile and hence the embedded value of the company.

[1/2 mark for each relevant point]

[Max 8]

vi)

Following could be the likely reasons:

- a. For solvency purposes, the bonds are expected to be *held till maturity* and any interest rates movement should not impact the assets valuation.
- b. However, for liability valuation, the Assets Liabilities and Solvency Margin regulation mandates that the insurer should consider the backing assets return along with the new money exposure in future while deriving the valuation rate of interest. Hence, the liabilities valuation for solvency purposes are sensitive to movement in interest rates in economy up to the extent of future new money investments from the existing business i.e. future renewal premiums, coupons and maturity proceeds from the existing assets holding.
- c. Hence, the quantum of exposure to new money either because of future renewal premiums (which would be higher for regular pay products compared to limited pay or single pay products) or coupons / maturity proceeds from the existing assets held (which would be higher in case the assets and liabilities are not matched) is one of the likely reason for sensitivity of solvency position to interest rate changes.
- d. Further, with movement in interest rates, the MAD applicable on Valuation Rate of Interest would also changes because as per the APS7, the MAD applicable on interest rate parameter is basis the current 10-year G-Sec yield. Hence, the changes in MAD position will impact the liability side with no negating impact on assets side.
- e. For linked business, the fund value is valued at mark-to-market basis and hence any interest rate movements in economy would impact the fund value of linked business. With that, the Required Solvency Margin (RSM) computation would also change as the RSM is calculated basis factor approach on reserves and Sum at Risk. Further, there will not be any corresponding change in the Available Solvency Margin (ASM) of the

company as the ASM considers Held Till Maturity model for existing assets. Higher the product mix towards linked business, higher would be the volatility of solvency ratio to interest rate movements.

- f. As per the 'Preparation of financial Statements and Auditor's Report' regulations, ARA regulations and ALSM regulations, companies dealing in actively traded derivative instruments would have to value the derivative assets at lower of book value and market value for solvency purposes. In case of interest rate up scenario, the market value of the interest rate derivative asset would be lower and that would impact the assets valuation to be considered for solvency purpose with no negating impact on liability side. This would make the solvency position of the company sensitive to interest rate movements in the economy.

[1/2 mark for each relevant point]

[Max 5]

vii) Measure to reduce the interest rate sensitivity are as follows:

a) For solvency

- The sensitivity to interest rate would reduce with reduction in future new money exposure of the company, which could be done through Asset Liability matching. Further, single premium or limited pay products have lower exposure to future new money compared to regular pay products.
- Minimizing the guarantees in the products specifically under linked and participating products would help in adjusting liabilities correspond to changes in assets to keep the solvency position immune.
- Another mitigating measure could be to enter into interest rates hedge agreements wherein the returns on future investable will get locked and any likely movement in interest rates in future would not impact the portfolio yield computation.
- The estimated level of new money assumed in the setting of the MAD calculations, will have an impact of the sensitivity of the solvency.
- Trading in interest rate derivative assets which are not-actively traded in market instead of trading in actively traded derivative instruments would help because as per the regulations, the non-actively traded derivative assets are valued at cost and hence any changes in the interest rate won't impact the assets valuation and hence solvency of the company.
- Furthermore, the interest yield curve in the economy could change with different magnitudes and direction for different durations and hence it is impossible to immunize the solvency of the company to any change in the shape or magnitude of the yield curve.
- The CFO's proposal to increase the equity proportion in Participating fund would have an impact on the interest rate sensitivity of the company's solvency. With reduced proportion in bonds, the assets become less sensitive to changes in interest rates, whereas the corresponding impact on liability side would depend on the correlation assumed between the interest rate and equity return while estimating the valuation rate of interest.

- Higher correlation assumption will make the liability side more volatile and hence the solvency position of the company will be volatile. Whereas, on assumption of insignificant correlation, the solvency would be less sensitive to interest rate movements

[Max 5]

b) For Embedded Value

- With changes in the yield curve, the value of assets and best estimate liabilities would change as both assets and liabilities are valued at market consistent basis.
- In case the insurance company's assets and liabilities are perfectly matched, i.e. both the duration and amount are matched then the company is immune to changes in interest rate movements, however, this is not achievable because of the following reasons:
 1. The assets of longer duration are not available compared to the duration of the liabilities of a life insurance company.
 2. The amount of assets held by insurance company are generally higher than the best estimate liability as mandated by the ALSM regulations.
- So, effectively the duration of assets of a life insurance company is lower compared to the duration of liabilities i.e. with 1% change in the interest rate the value of assets would move by less percentage compared to movement in liabilities. But, the amount of assets held by insurance companies are higher compared to liabilities. Hence, the company could implement a strategy wherein the lower percentage impact on higher asset base could negate the higher percentage impact on lower liability base resulting in overall negating impact on economic surplus.
- Having said that, like for solvency immunization to interest rate movements, the embedded value of the company can't be immunize to any change in shape and magnitude of the yield curve. Hence, the ultimate objective should be to keep the movement in the EV within the tolerance limit for any magnitude of interest rate changes at any duration.
- The CFO's proposal to increase equity investments in Participating fund would reduce the company's embedded value sensitivity to interest rate changes. However, it would increase the equity market sensitivity.

[1 mark for each relevant point]

[Max 4]

viii)

Key considerations while setting the assumptions and methodology for estimating the valuation rate of interest are as follows:

- a. The valuation rate of interest is used to discount the future cashflows corresponds to existing policies for the estimation of reserves to be shown in Balance Sheet. The valuation rate of interest should reflect the expected return the Company expects to earn on the backing assets along with the return expected from investments expected in future on prudent basis.
- b. For that, the cash flows correspond to existing assets and existing policy liabilities need to be projected in future. The coupons and maturity proceeds from the existing assets projection along with the future renewal premium from the existing business would

- help meeting the existing policy payouts. The need for investment in future new money will get identified through cashflow matching exercise.
- c. Further, future new money return need to be assumed basis the projection of yield curves that could prevail in future.
 - d. As per the ALSM regulations, Valuation rate of interest, to be used by Appointed Actuary for the calculation of the present value of policy cash flows shall not be higher than the rates of interest, determined from prudent assessment of the yields from existing assets attributable to blocks of life insurance business, and the yields which the insurer is expected to obtain from the sums to be invested in the future, and such assessment shall take into account-
 - the composition of assets supporting the liabilities, expected cash flows from the investments on hand, the cash flows from the block of policies to be valued, the likely future investment conditions and the reinvestment and disinvestment strategy to be employed in dealing with the future net cash flows;
 - the risks associated with investment in regard to receipt of income on such investment or repayment of principal;
 - the expenses associated with the investment functions of the insurer;
 - e. VROI shall not be higher than, for the calculation of present value of policy cash flows in respect of a particular category of contracts, the yields on assets maintained for the purpose of such category of contracts;
 - f. In respect of non-participating business, shall recognize the risk of decline in the future interest rates;
 - g. In respect of participating business, shall be based on the assumption (with regard to future investment conditions), that the scale of future bonuses used in the valuation is consistent with the valuation rate of interest.
 - h. An Appointed Actuary shall take into consideration the following,
 - The expected portfolio yield shall be adjusted by an appropriate Margin for Adverse Deviations (MAD), the level of MAD being dependent on the degree of confidence in the expected level, and such MAD in each parameter shall be based on the Actuarial Practice Standards / Guidance Notes issued by the Institute of Actuaries of India, with the concurrence of the Authority
 - The values used for the various valuation parameters should be consistent among themselves.

[½ marks for each point] **[Max 6]**
[50 Marks]
